

Bank Account Management System

Title: Bank Account Details using Classes and Objects

Objective/Aim:

To write a Java program to implement a simple Bank Account Management System using a class, constructor, methods, deposit, withdrawal, interest calculation, and account details display[cite: 7].

System Requirements :

Hardware

- Standard PC/Laptop[cite: 1]
- Intel Core i3/i5 processor or equivalent[cite: 1, 7]
- Minimum 4 GB / 8 GB RAM[cite: 1, 7]
- Minimum 500 MB free storage[cite: 1]
- Keyboard and Monitor[cite: 1]

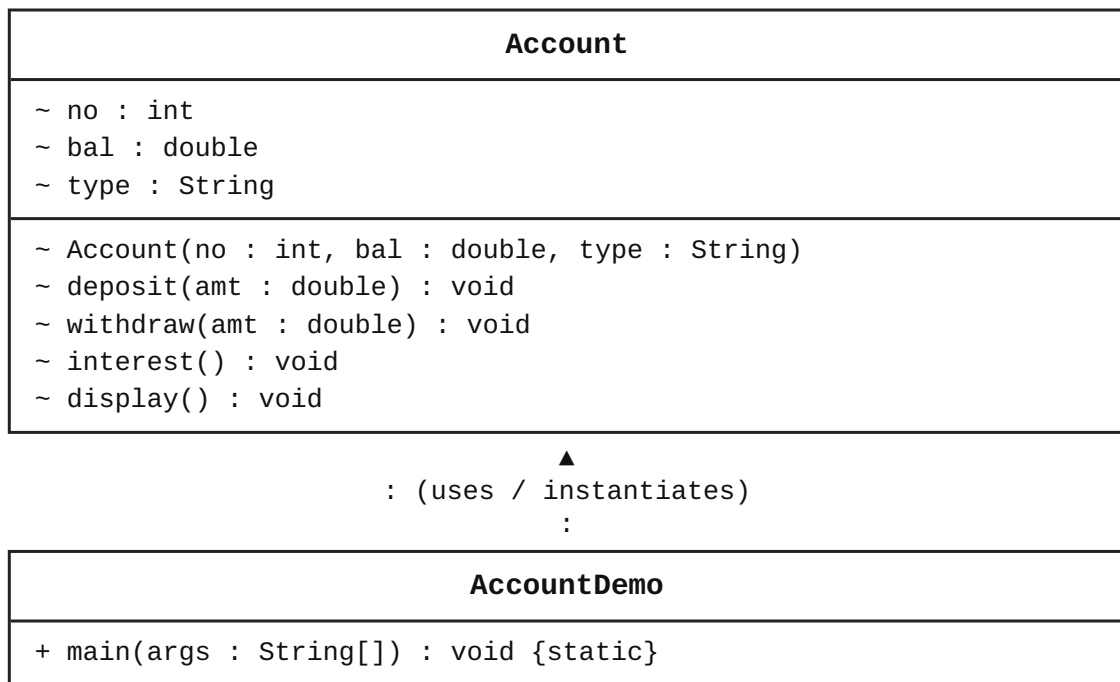
Software

- Operating System: Windows/Linux/macOS[cite: 1, 7]
- Java Development Kit (JDK 17 or JDK 21)[cite: 1, 7]
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code[cite: 1, 7]
- Command Prompt/Terminal[cite: 1]

Algorithm / Logic Description:

1. Create a class named `Account` with attributes `no`, `bal`, and `type`[cite: 7].
2. Create a parameterized constructor to initialize account details[cite: 7].
3. Define `deposit(double amt)` to add the deposited amount to the balance and print the value[cite: 7].
4. Define `withdraw(double amt)` to check balance and deduct funds when sufficient, otherwise display an insufficient balance message[cite: 7].
5. Define `interest()` to calculate and display interest at 4% of the current balance[cite: 7].
6. Define `display()` to print the account number, account type, and balance[cite: 7].
7. Create the `AccountDemo` class containing the `main()` method[cite: 7].
8. Instantiate an `Account` object with account number 1001, balance 10000, and type "Savings"[cite: 7].
9. Call `display()`, `deposit()`, `withdraw()`, `interest()`, and `display()` to perform all banking operations[cite: 7].
10. Stop the program.

UML class Diagram



Relationship: AccountDemo creates objects and invokes operations of the Account class[cite: 7].

Source Code:

```
class Account {
    int no;
    double bal;
    String type;

    Account(int no, double bal, String type) {
        this.no = no;
        this.bal = bal;
        this.type = type;
    }

    void deposit(double amt) {
        bal += amt;
        System.out.println("Deposited: " + amt);
    }

    void withdraw(double amt) {
        if (amt <= bal) {
            bal -= amt;
            System.out.println("Withdrawn: " + amt);
        } else {
            System.out.println("Insufficient Balance");
        }
    }

    void interest() {
        System.out.println("Interest: " + (bal * 0.04));
    }
}
```

```

    }

    void display() {
        System.out.println("No: " + no);
        System.out.println("Type: " + type);
        System.out.println("Balance: " + bal);
    }
}

public class AccountDemo {
    public static void main(String[] args) {
        Account a = new Account(1001, 10000, "Savings");

        a.display();
        a.deposit(2000);
        a.withdraw(3000);
        a.interest();
        a.display();
    }
}

```

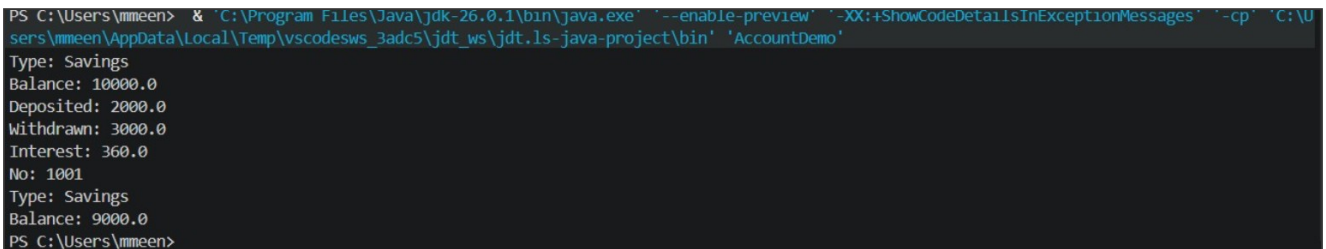
Compilation & Execution Commands:

Save file as: AccountDemo.java

Compilation: javac AccountDemo.java

Execution: java AccountDemo

Output Screenshot:



```

PS C:\Users\mmeeen> & "C:\Program Files\Java\jdk-26.0.1\bin\java.exe" --enable-preview --XX:+ShowCodeDetailsInExceptionMessages -cp "C:\U
sers\mmeeen\AppData\Local\Temp\vscodesws_3adc5\jdt_ws\jdt.ls-java-project\bin" "AccountDemo"
Type: Savings
Balance: 10000.0
Deposited: 2000.0
Withdrawn: 3000.0
Interest: 360.0
No: 1001
Type: Savings
Balance: 9000.0
PS C:\Users\mmeeen>

```

Result:

The Bank Account Management System program was successfully implemented in Java using classes and objects[cite: 7]. The program demonstrates the use of a parameterized constructor, object creation, data members, and methods to deposit, withdraw, compute interest, and display state transitions of a bank account[cite: 7].